# Shrub Hub
## USER GUIDE

## Overview

This document contains the necessary information to use the Shrub Hub, a low cost, easy to build digital switch interface.

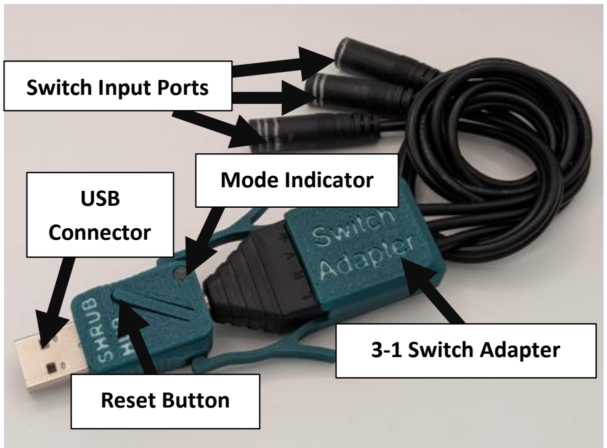

## Introduction

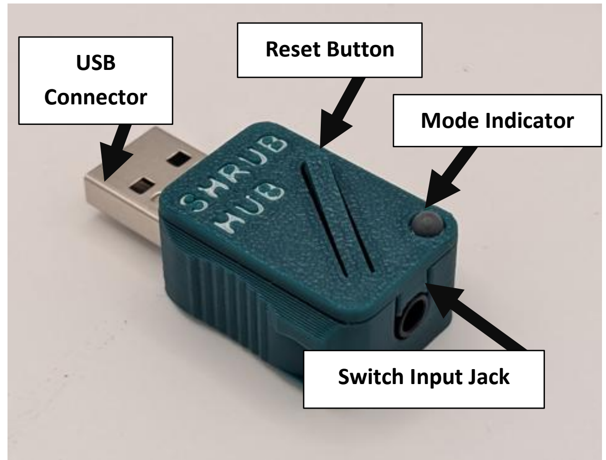
The Shrub Hub is a digital USB switch interface that allows a user to connect up to three 3.5 mm assistive switches to a digital device like a computer or mobile phone. It has five operating modes, and can send keystrokes, mouse clicks, and media control commands, with five outputs per mode. The Shrub Hub is powered from the device and does not require external power. The Shrub Hub uses standard USB Human Interface Device (HID) drivers and does not require the installation of any software.

It is designed to provide digital access to users using assistive switches, and to be low cost and very easy to build.

# Contents

## Features




**Switch Input Ports:** The switch ports where 3.5 mm assistive switches can be connected.

**USB Connector:** The USB Connector is a USB-A plug that is inserted into a USB-A port on the Host Device.

**Reset Button:** The reset button of the microcontroller.

**Mode Indicator:** A multicolour LED that indicates the current operating mode.

**3-1 Switch Adapter:** An optional component that can be built and attached to the Shrub Hub. It can connect to up to three switches and allows the user to send inputs to the Shrub Hub using all three, instead of just one if the Switch Adapter is not used.

## Specifications

| Item | Shrub Hub | Shrub Hub + Switch Adapter |
|---|---|---|
| **Size (Length x Width x Height) [mm]** | 40 x 20 x 10 | 100 x 40 x 10 |
| **Mass [g]** | 8 | 48 |
| **Assistive Switch Inputs** | 1 | 3 |
| **Configurable Outputs per Mode** | 1 | 5 |

## Compatibility

**Assistive Switches:** The Shrub Hub is compatible with up to 3 assistive switches that have a 3.5 mm plug.

**Host Device:** The Shrub Hub is compatible with any digital host device that accepts USB HID inputs and has a USB-A port, such as a computer or mobile device. If the connected device does not have a USB-A port, an adapter will be needed. If used with an iOS device, it will need a separate power source as described in the iOS use section.

# Usage

The Shrub Hub is a digital switch interface that generates outputs to a host device (e.g., computer, smartphone) when the attached assistive switches are activated. The initial setup involves attaching 1-3 assistive switches to the Shrub Hub, or via the Switch Adapter, then connecting the Shrub Hub to the host device. During use, the Shrub Hub will generate USB HID outputs such as keystrokes, mouse clicks, or media control inputs depending on the operating mode when the connected assistive switches are activated. The current operating mode is indicated by the colour of the indicator light. The operating mode can be changed by performing a long press (holding for longer than four seconds) on a connected assistive switch.

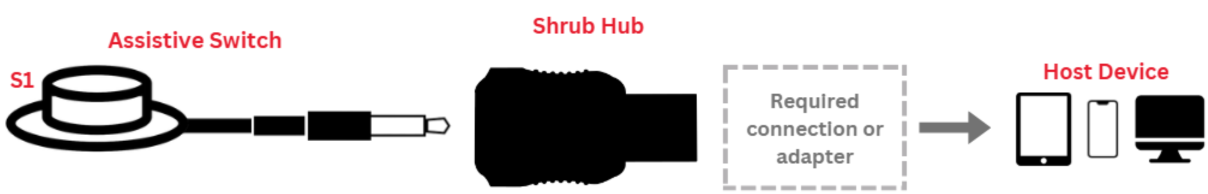
## Initial Setup

The Shrub Hub can be setup in two configurations: a Single Switch Configuration for use with a single assistive switch, or 1-3 Switch Configuration for use with 1, 2, or 3 assistive switches. The configurations differ by whether the 1-3 Switch Adapter is required.

| | Configuration | |
|---|---|---|
| | Single Switch Configuration | 1-3 Switch Configuration |
| **Hardware** | Shrub Hub | Shrub Hub + Switch Adapter |
| **Assistive Switches** | 1 | 1, 2, or 3 |
| **Outputs Per Mode** | 1 | 5 |

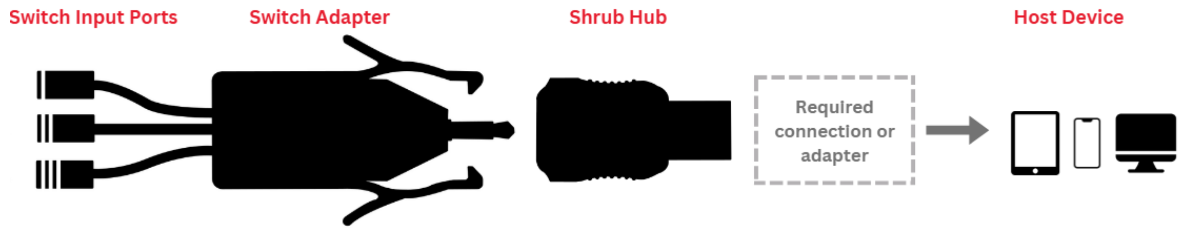
### Single Switch Configuration

If a user is only going to use a single assistive switch, the switch can be plugged directly into the Shrub Hub. When using a single switch, the Shrub Hub reads the switch as Switch 1+3 or Switch 1+2+3 (depending on the type of cable) on Table 1 found below, which functions the same as Switch 1.

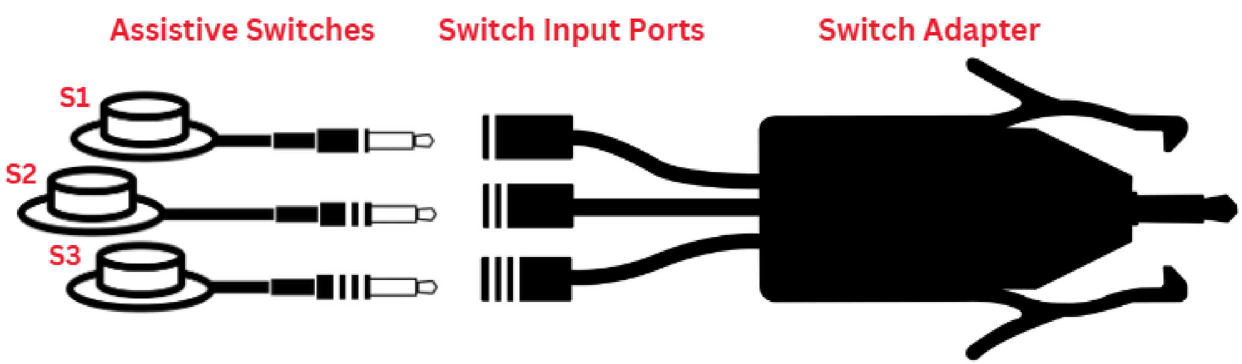


## 1-3 Switches

When setting up the Shrub Hub with more than one switch, the first step is to connect the Switch Adapter to the Shrub Hub.




After plugging in the Switch Adapter, plug Switch 1 into Switch Input Port 1, Switch 2 into Switch Input Port 2, and Switch 3 into Switch Input Port 3. The cables may have been marked by the maker of the device, but if not, they should be marked on the first use to make future setups easier. The easiest way to mark them will be to plug in the device and press each button. Refer to Table 1 on the last page of the guide for which button should output which keystroke. Using the switch outputs, you can determine which switch is which and label the splitter cables as such.



# Regular Use

## Connecting to Host Device

After plugging in the switch or switches to the Shrub Hub, plug the Hub into a USB-A port on the host device.

- For host devices with a USB-A port (e.g., computer), the Shrub Hub can be connected directly or connected via a USB-A extension cable.
- For host devices without a USB port other than USB-A, use a suitable adapter.
- For iOS devices, a powered USB Hub or powered USB splitter is required to connect the Shrub Hub to the device.

Once connected and powered, a light on the Shrub Hub will turn on and indicate the Operating Mode, at which point the Hub is ready to use.

## Generating Output

Once the Hub is ready to use, activating one or more of the connected switches will generate the corresponding output based on the current Operating Mode. The output will be produced when the switch is released.

### *Single Switch Activation*

### *Multiple Switch Activation (Chording)*

If two or more switches are activated simultaneously, a different output will be produced.

While the Shrub Hub has three switch inputs when using the Switch Adapter, it can send up to five different outputs using chording, which is pressing multiple switches at the same time. A chorded input is activated on the Shrub Hub when two switches are released within 0.5 seconds of each other.

The Shrub Hub currently only has chorded outputs on switches 1+2 and switches 2+3; switches 1+3 and switches 1+2+3 perform the same output as switch 1 as plugging a single switch into the Shrub Hub directly instead of using the Switch Adapter will send the same signal as using the Switch Adapter with those chorded inputs.

See the below table for the full list of outputs.

## Operating Modes and Outputs

The Shrub Hub has different outputs depending on the operating mode. The different operating modes are as follows:

Operating Mode 1 – This mode has the keystrokes Enter, Space, Tab, Backspace, and Escape. These keystrokes can be used to navigate a web browser, or for switch scanning on a mobile device.

Operating Mode 2 – This mode has the first five function keys, F1-F5. These keystrokes can be used for switch scanning on a mobile device

Operating Mode 3 – This mode has left, right, and middle mouse clicks. This mode can be used in tandem with a USB joystick to move the mouse, then use the Shrub Hub to click on items.

Operating Mode 4 – This mode has the first five unused function keys. They are unused in most programs, and can be used with programs such as AutoHotkey to trigger preprogrammed actions or scripted programs.

Operating Mode 5 – This mode has the media control commands Play/Pause, Volume Up, Volume Down, Next Track, and Previous Track. This allows the Shrub Hub to be used to control the media playing on the connected device.

**Table 1: Default outputs for each Operating Mode in multiple switch mode**

| Input | | Output | | | | |
|---|---|---|---|---|---|---|
| Activation | Input Switch | Mode 1 (Teal) | Mode 2 (Purple) | Mode 3 (Yellow) | Mode 4 (Green) | Mode 5 (Orange) |
| One Switch | Switch 1 | Enter | F1 | Left Click | F13 | Play/Pause |
| | Switch 2 | Space | F2 | Right Click | F14 | Volume Up |
| | Switch 3 | Tab | F3 | Middle Click | F15 | Volume Down |
| Two Switch | Switch 1+2 | Backspace | F4 | Left Click | F16 | Next Track |
| | Switch 2+3 | Escape | F5 | Right Click | F17 | Previous Track |
| | Switch 1+3 | Enter | F1 | Left Click | F13 | Play/Pause |
| Three Switch | Switch 1+2+3 | Enter | F1 | Left Click | F13 | Play/Pause |

**Table 2: Default outputs for each Operating Mode in single switch mode**

| Input | | Output | | | | |
|---|---|---|---|---|---|---|
| Activation | Input Switch | Mode 1 (Teal) | Mode 2 (Purple) | Mode 3 (Yellow) | Mode 4 (Green) | Mode 5 (Orange) |
| One Switch | Switch 1 | Enter | F1 | Left Click | F13 | Play/Pause |

### Changing Operating Mode

The current Operating Mode is indicated by the colour of the Mode Indicator on the Shrub Hub. The Operating Mode can be changed so that the Shrub Hub generates a different set of outputs.

To change Operating Modes, hold any switch for more than four (4) seconds. The Mode Indicator will turn off when the switch has been pressed and turn back on with the same colour once the mode change time has been exceeded (i.e., when the switch has been held for at least four seconds). Once the switch has been released, the mode colour light will change to the colour of the next mode and the Shrub Hub will not send an output. The mode cannot change more than once per switch press and will not change until the switch is released.

### Changing Outputs

The outputs of each mode can be changed to better suit the needs of the user. The instructions for how to do this can be found in the Changing Outputs Guide in the Shrub Hub documentation folder. This requires opening and modifying the code, and should only be done by someone who is confident in their ability.

## Storage

Disconnect the assistive switches by gripping the plug and jack and gently pulling apart.

When not in use, the Shrub Hub should be stored in a cool place out of direct sunlight.

# Care

The Shrub Hub enclosure is made of 3D printed plastic. Exposure to high heat may cause warping and/or negatively affect function. Extended exposure to sunlight will also weaken the plastic on the device.

The Shrub Hub contains electronics and is not waterproof. If the device becomes wet, make sure it is disconnected and do not use it until it has completely dried. It may help to open the enclosures to speed up drying and ensure it has completely dried.

## Cleaning

The Shrub Hub can be wiped with a damp cloth while unplugged from the device.

## End of Life / Disposal

PLA filament may be industrially compostable in your area. Check with your waste management company if PLA can be composted or must be thrown in the garbage.

Disassemble the Shrub Hub and separate out the recyclable and compostable components, and those that must be thrown out. Electronics and batteries should be disposed of following your local waste management guidelines.

# Changing the Outputs

For some users, the provided switch outputs may not meet their needs. It is possible to edit the outputs of the Shrub Hub, but this involves editing the code stored on the device. This is not an easy process, and should only be done if you feel confident and while carefully following the instructions.

Detailed instructions on how to change settings on the Shrub Hub can be found in the Shrub Hub Changing Outputs Guide, found here, or at the following QR code.

**Figure 1: QR Code Link to the Changing Outputs Guide**

# Troubleshooting

This section helps you identify and fix common issues when using the Shrub Hub. If the problem persists after trying the steps below, see the **Support and Resources** section.

## Quick Checks

- Make sure the Shrub Hub is securely connected to a USB port on host device.
- Verify that the cable of each assistive switch is firmly attached to the corresponding input port.
- Verify that the Mode Indicator is lit and showing the color of the intended operating mode.

## Common Problems and Solutions

| | Problem, Issue, or Concern | Cause | Remedy |
|---|---|---|---|
| 1.1 | Nothing happens when switch is activated | Switch isn't properly connected | Confirm that each switch is firmly connected to the intended input port. |
| 1.2 | | Switch isn't working | Confirm that the switch is working with a different device or switch tester. |
| 1.3 | | A non-noticeable output is generated from a different operating mode than expected | Check the Mode Indicator to ensure that the Shrub Hub is in the desired operating mode. If not, Change Operating Mode. |
| 1.4 | | Loose or disconnected Switch Adapter | Confirm that the Switch Adapter is securely connected to the Shrub Hub. |
| 2.1 | Unexpected output is generated when switch is activated | Different switch activated | Confirm each switch is connected to the intended input port |
| 2.2 | | Different operating mode than expected | Check the Mode Indicator to ensure that the Shrub Hub is in the desired operating mode. If not, Change Operating Mode. |
| 3.1 | Mode Indicator is blinking yellow | Host device does not support USB HID | Connect Shrub Hub to a device that supports USB HID |
| 3.2 | | Shrub Hub connected to USB Power Adapter | Connect Shrub Hub to a device that supports USB HID |
| 4.1 | Mode Indicator is blinking red | The Shrub Hub code is not compiling | Revert the code to the latest version found on the Shrub Hub GitHub. |
| 5.1 | The Shrub Hub freezes and does not respond to button presses | Shorted cables in Switch Adapter | Open the Switch Adapter enclosure and ensure there is no short between two of the cables. |

# Support and Resources

## Getting Help

The device referred to in this guide is an open-source assistive technology and is supported by a combination of documentation and community support.

If you need help using this device, please start by reviewing the documentation, including the Troubleshooting section. If you are still having trouble, you can reach out for support in several ways:

1. Email the Makers Making Change Team at info@makersmakingchange.com.
2. Reach out to Makers Making Change on social media.

When asking for help, it is helpful to include:

- The name of the device
- A brief description of the issue
- What you were trying to do when the problem occurred
- Photos or videos, if available

## Project Page & Documentation

The project page contains the most up-to-date information about this device, including documentation, design files, and known issues.

- Project page: https://github.com/makersmakingchange/Shrub-Hub

- User Guide: https://github.com/makersmakingchange/Shrub-Hub/blob/main/Documentation/Shrub_Hub_User_Guide.pdf

**Figure 2: QR code link to project files**

## Reporting Issues or Suggesting Improvements

If you want to report a problem with the Shrub Hub, or suggest an improvement, you can get in touch in the following ways:

- Leave an issue on the GitHub repository, found at
    - https://github.com/makersmakingchange/Shrub-Hub/issues
- Send us an email at info@makersmakingchange.com

## Open Source & Licensing

The device referred to in this guide is open-source assistive technology. This means the design files and documentation are publicly available and may be used, modified, and shared under the terms of the project's license.

Open-source devices are often built and adapted by volunteers. While care is taken to provide accurate documentation, performance cannot be guaranteed in every situation. Always use your judgment and consult a professional if you have questions about suitability or safety.